ML Assignment 3

UNIVERSITY OF HARTFORD

Multiple Linear Regression

with Advanced Extensions

Building a Multiple Linear Regression Model to predict house price ranges.

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1 Objective

In this assignment, a linear regression model based on multiple features was built, evaluated, and ehanced by polynomial features and regularization. Procedures included:

- Created basic linear regression model
- Extended basic model with Polynomial Features
- Applied Ridge Regression for regularization
- Visualized the results and analyzed indiviual feature impacts

2 Data Set Overview and Cleaning

Table 1: Summary Statistics of House Prices

attribute	mean	std	min	25%	50%	75%	max	data_type	num_null	num_unique
Square_Footage	2815.42	1255.51	503.00	1749.50	2862.50	3849.50	4999.00	int64	0	894
Num_Bedrooms	2.99	1.43	1.00	2.00	3.00	4.00	5.00	int64	0	5
Num_Bathrooms	1.97	0.82	1.00	1.00	2.00	3.00	3.00	int64	0	3
Year_Built	1986.55	20.63	1950.00	1969.00	1986.00	2004.25	2022.00	int64	0	73
Lot_Size	2.78	1.30	0.51	1.67	2.81	3.92	4.99	float64	0	1000
Garage_Size	1.02	0.81	0.00	0.00	1.00	2.00	2.00	int64	0	3
Neighborhood_Quality	5.62	2.89	1.00	3.00	6.00	8.00	10.00	int64	0	10
House_Price	618861.02	253568.06	111626.85	401648.23	628267.29	827141.28	1108236.84	float64	0	1000

The data set used for training and testing of the multiple linear regression model consists of 1000 records each with 8 features, representing unique properties and their particular characteristics. The data is very clean with few issues to be addressed. There are no explicit null values. The number of unique values, and the data types are mostly consistent with each other, indicating limited need to impute or transform any of the features at this stage.

One potential exception for cleaning is the "Garage_Size" feature where its minimum value and 25th percentile are both zero. This value is implicitly a null value, indicating that the property does not have a garage. In context of the report, this value indicates that at least 25% of the data records do not have a garage. The other values from this size are "1" and "2", based on the fact that they are the median and max values, respectively, and there are only a total of three unique values in the feature. These values either quantify the number of distinct garages or the number of bays in the garage, but it is not clear which is the case. In either case "0" indicates no garage. This means that "Garage_Size" could be separated into two variables- one that indicates whether the property has a garage and the other that quantifies the number of garages/bays. However, since the current feature unambiguously encodes this information as is, the feature is left in its original state.

All of the features are accurately characterized by the data type assigned to it by Python's PANDAS library on import, with the exception of the "Neighborhood_Quality" attribute whose numeric values are proxies for ordinal quantitative grades of the neighborhood in which each property is found. For ordinal data, there is no meaningful magnitude inherent in the values regardless of whether or not the values are numbers or character strings. Because of this, the feature can be converted to a

categorical type that will eventually be encoded with rank in preparation for fitting the linear model. This would ensure that the values will not be treated as interval data types, in spite of the values being numbers. However, unless a number-valued ordinal feature's values have an inverse rank, it is best to leave it as is. In this case the rank is directly aligned with the set of values the feature can take. The values 1-10 represent the rating of neighborhoods lowest to highest already (kaggle). When creating ordinal data types, the linear regression algorithm will eventually need it to be converted back to numbers to satisfy the needs of the algorithm and therefore it is unproductive to update the feature type simply for linear regression model training.

Ultimately, cleaning was not required for this dataset. The features and their values are already optimal for training linear regression models.

3 Data Preparation and Model Training

Three different approaches for generating multiple linear regression models were carried out and the results of their individual techniques were compared. This included building a basic linear regression model as well as models that include polynomial transformations of numeric features in order to enhance the performance of the trained model and regularization of the polynomial model to increase its precision. Evaluative statistics were collected for each model technique to assess the their relative effectiveness. Simple linear models were built that corresponded to the techniques used in for the multiple regression models, providing insight into the amount of variance that each individual feature accounted for in the data.

3.1 Basic Linear Model

The basic linear model utilized all available features in the source data without consideration for the individual contributions of each feature to the overall variance in the data explained by the model. In addition to the evaluative statistics and corresponding simple linear models, feature performance was evaluated to provide insight into what features had the greatest impact on the multiple regression model and how that corresponded to the effectiveness of the simple linear models.

3.2 Polynomial Features Model

Next multiple linear regression models were produced. The numeric features were were transformed into polynomial features that distribute the information contained in a single feature across multiple features. This distribution of information allows for non-linear relationships between features to be captured in the model which is reflected in its predictions. These relationships are established through raising the degree on the values of each independent variable as separate features as well as defining features that include the product between features. This significantly augments the dimensions of the training and testing data and can potentially cause overfitting when a basic model already performs well but is a reliable method for improving linear models generally. Fitting the simple models using the same polynomial degrees of the multiple regression models, clarifies how the predictive curve becomes more responsive to patterns in the data.

3.3 Ridge Regression Polynomial Features Model

Finally, multiple linear regression models were produced using the ridge regression technique. This is a regularization technique for increasing precision on a module while sometimes introducing bias. Often this bias is very samll, making the ridge technique's ability to increase precision worth the trade off. These models each used the same degrees of polynomial enhancement as was used in the prior models. In the context of polynomial features regularization can mitigate some of the multicollinearity that is introduced by the dimensional expansion of the source data.

4 Results and Discussion

Table 1: Model Comparison

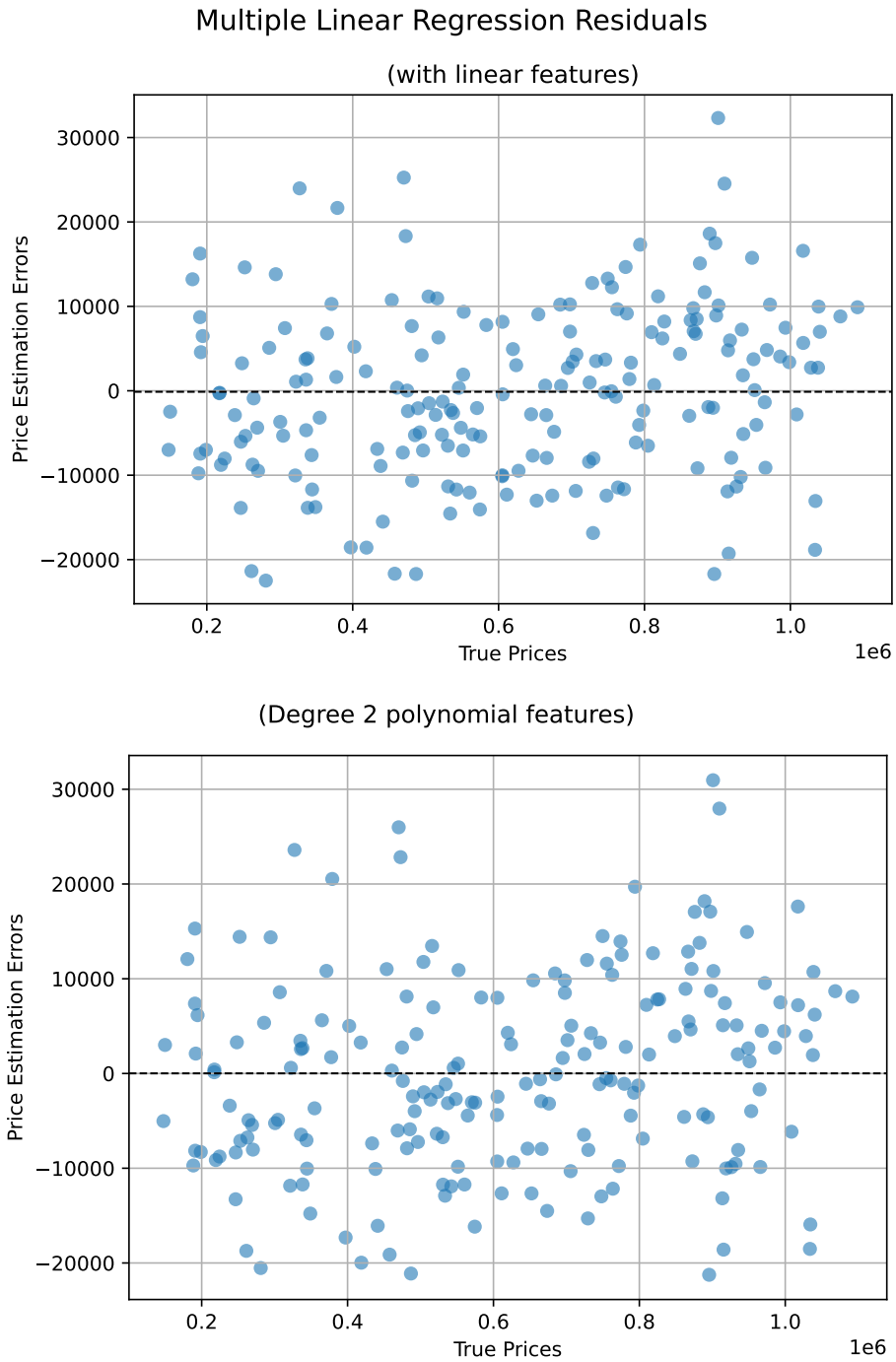
model	mse	R^2	intercept
linear	101434798.505668	0.998426	-2010567.420270
poly_deg5	138936844.541452	0.997845	-1234636.245764
poly_deg2	103781419.476088	0.998390	247486.172713
ridgePoly_deg5	1452124133.422956	0.977472	157520255.570688
ridgePoly_deg2	103544048.438783	0.998394	500815.526725

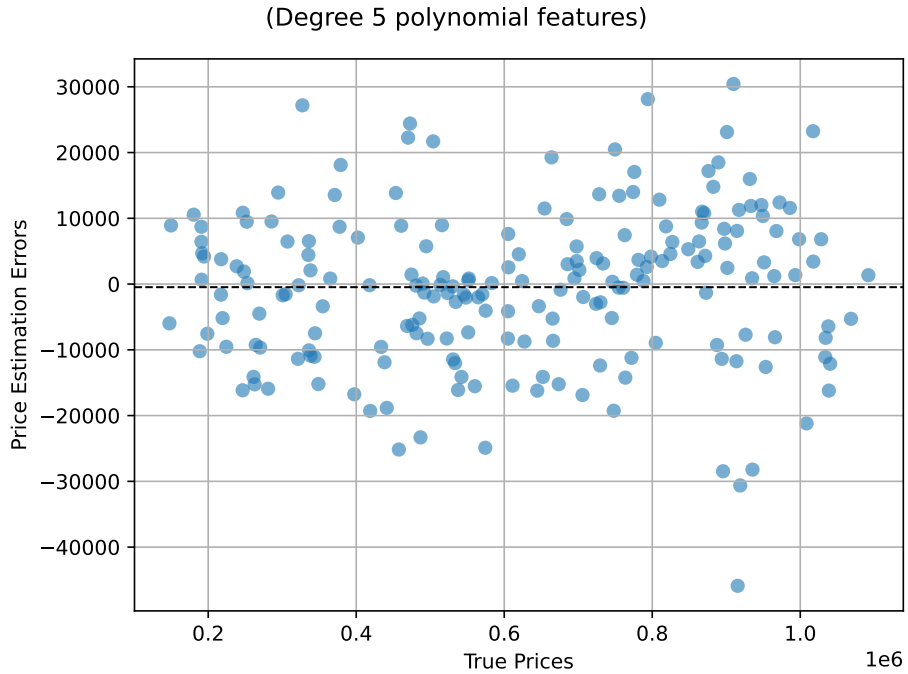
Five multiple linear regression models were fitted on the same testing and training data sets with each set consisting of the same data entities. This resulted in comparable performance. In fact, the linear model without polynomial factor enhancements and ridge regression performed the best (referred to as basic model going forward. We can see this in the R^2 and mean squared error (MSE) scores in Table 1 where there is lesser amount of cumulative error with a greater amount of variance in the data explained by the model than in the subsequent models. The seaming lack of improvement is likely due to the high level of performance that already exists in the basic model, leading to the increase in features through polynomial enhancement to result in slight overfitting. The degree of polynomial enhancement is impactful resulting in both a larger MSE score and a smaller R^2 both of which is the opposite of what is intended by polynomial enhancement. Ideally, these measures should reflect better performance as a result of the nonlinear patterns that are made more apparent. Of the two polynomial enhanced models, degree 2 performed better than degree 5 indicating that degree 5 provided too much complexity. There is a notable increase in the ridge model for polynomial degree of 5 with a magnitude increase by a factor of 10 in the MSE score, which corresponded with the lowest R^2 score.

There was significant fluctuation between intercept values. Here, the intercept has little interpretable meaning. It indicates the predicted house price values when all of the independent attributes are zero. which is outside the meaningful scope of the property characteristics considered here- for example, there are no houses you can purchase that have a nonexistent square footage and lot size. The intercept can be impactful on statistical testing for comparing linear models on the same features in different contexts, however. For example, if you were to want to compare the effect of zip code on house prices, you can compare linear models produced from the same attributes but with datasets separated by the zip code the properties were found in. Statistically significant differences in the intercepts of corresponding linear models can

indicate a meaningful difference in expected prices, even if the models have similar coefficient values. With this in mind, the intercept of output can play an important role likely meaning that analysis multiple linear models might impact what hyperparameters are used when enhancing linear models, so it seems important to consider which is best. In this case, it is assumed that the basic models intercept is the most effective since it has the best performance overall.

Figure 1: Multiple Linear Regression Residual Plots

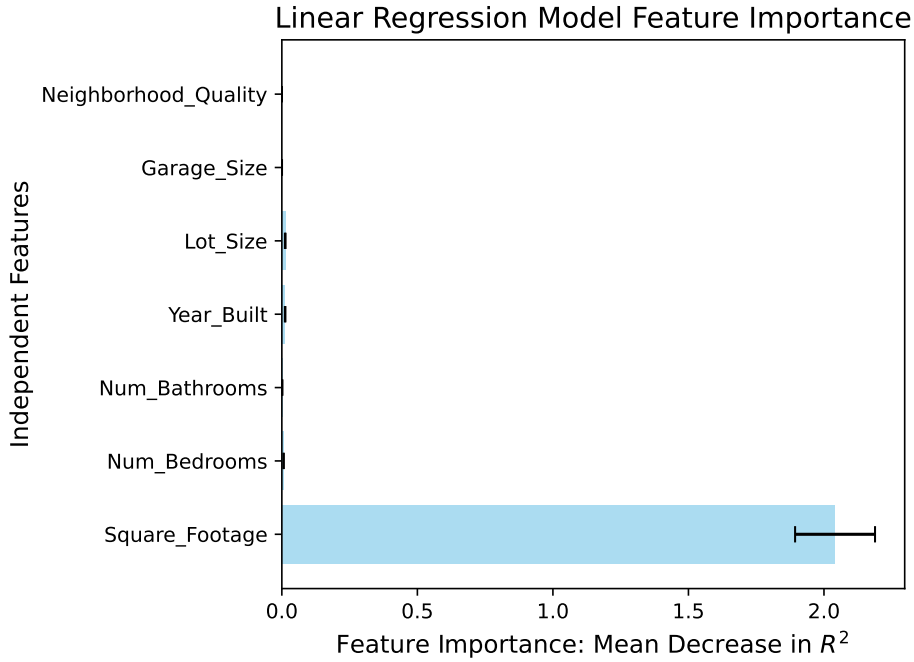




The three graphs in Figure 1 compare the average MSE between the basic and polynomial linear regression models. Where the MSE provides a sense of the overall precision in the model, its average can be used to clarify any bias. There is a slight definition of the residuals from the mean in the 5 degree polynomial model further indicating it having a slightly lower performance. The slight deviation persisted for the ridge model with the same polynomial features. This small deviation has little impact on the expected performance of the model, however.

Understanding that the basic model is the best model based on the performance of the model as well as lesser computational complexity warrants further exploration of its characteristics. The individual role of each factor can be seen in Figure 2.

Figure 2: Feature Importance Plot



The feature importance graph shows overwhelmingly that the "Square Footage" serves as the most important feature in the model. The feature importance algorithm works by shuffling the values of the feature being tested while holding the other feature values in fixed positions within the data matrix and determined the effect on the R^2 value. Plotting the mean decrease to the R^2 value indicates which features had the greatest role in the linear model. Here square footage had the greatest impact on the price of a house. Other impactful features included the lot size, year built and the number of bathrooms, but none of these had made a significant contribution overall. This is corroborated by the relationship between the fitted line of the simple regression model for square footage and the bi-variate behavior between the square footage and house price in data objects when compared to the other most important features. As seen in Figures 3-5, the plot of the simple model fit line for basic models and the bi-variate data points for the three most important predictors in the multi-linear model. The extent to which the line matches the variation in the predictor and predicted features in the simple model demonstrates the conceptual significance of feature importance significance.

Figure 3: "Square Footage" Simple Model vs Output Bi-variate Points

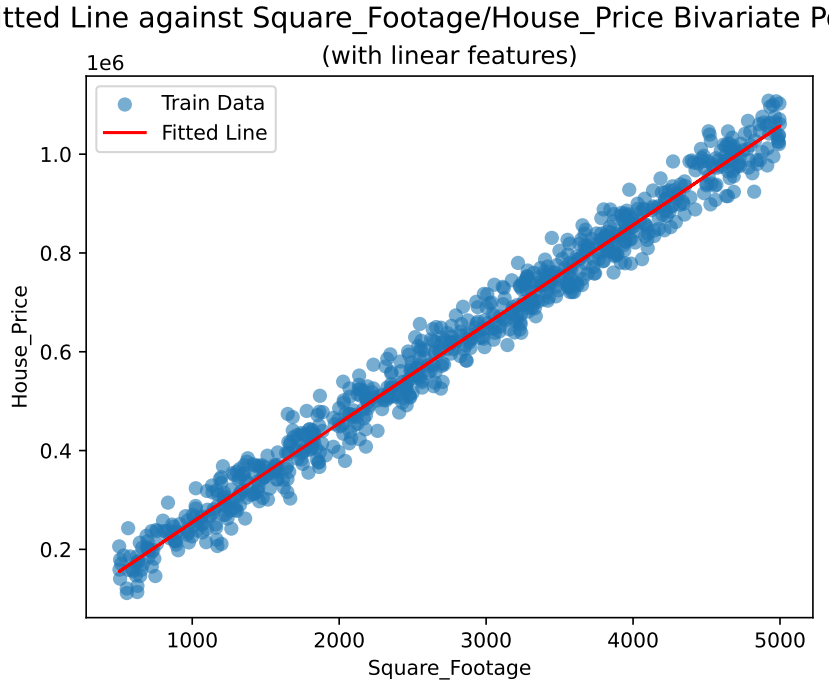


Figure 4: "Lot Size" Simple Model vs Output Bi-variate Points

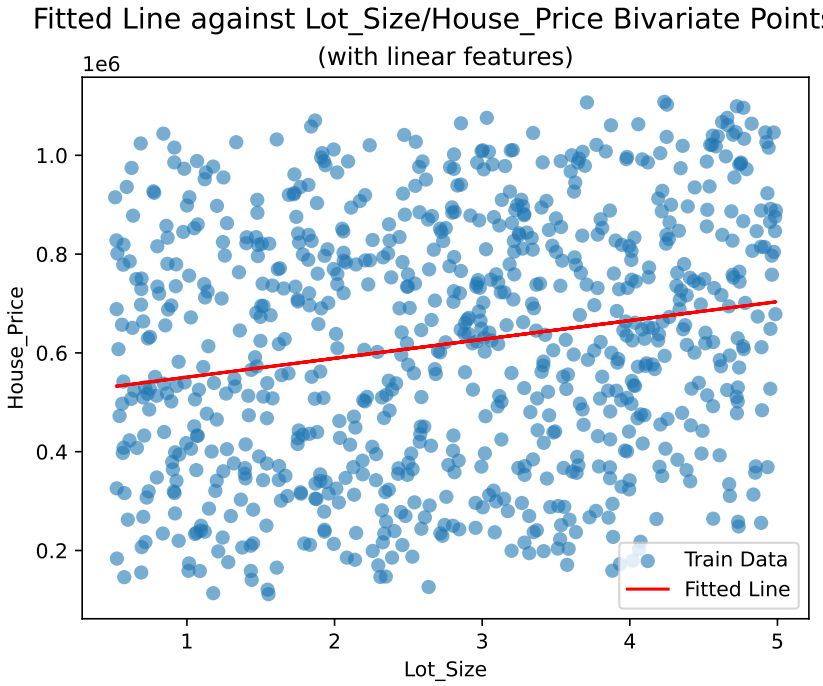
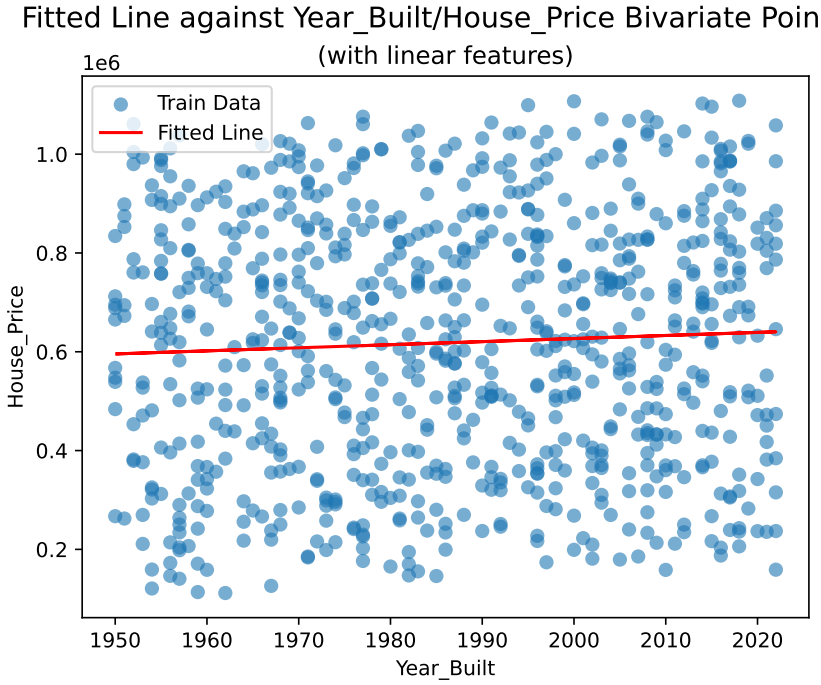


Figure 5: "Year Built" Simple Model vs Output Bi-variate Points



This trend in variation explained by the simple models perpetuated across the polynomial and ridge polynomial models as well, but with slightly different fitted lines reflecting the nonlinear relationships they seek to model (Figures 6 & 7). However, the fit line may show more change in slope in the data than actually exists. If this is the case, it would be consistent with the disparity seen between the MSE scores and between the R^2 scores in Table1. The difference in trend line can be seen in Figures 6 & 7 that show polynomial models against the bi-variate data.

Figure 6: "Square Footage" Degree 5 Polynomial Simple Model

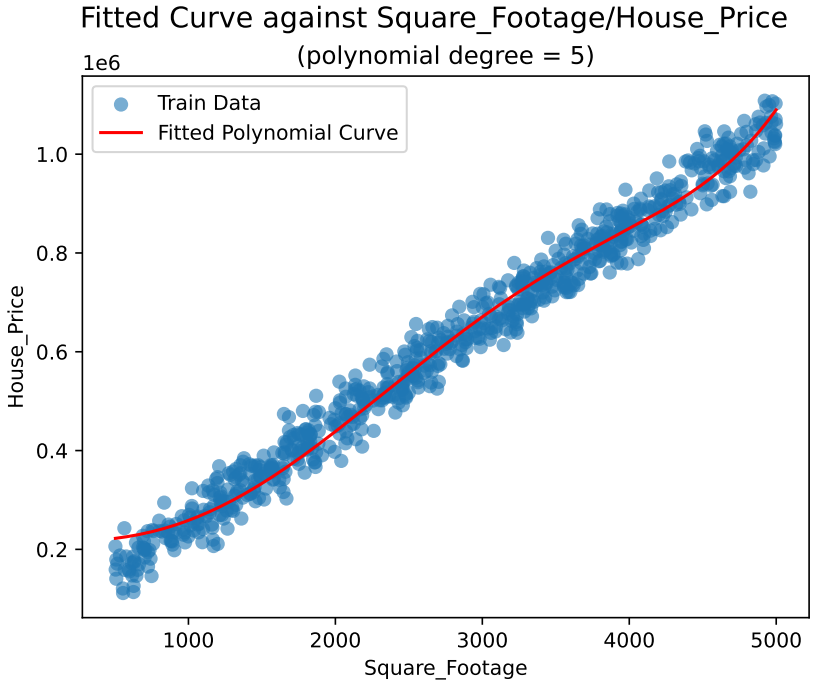
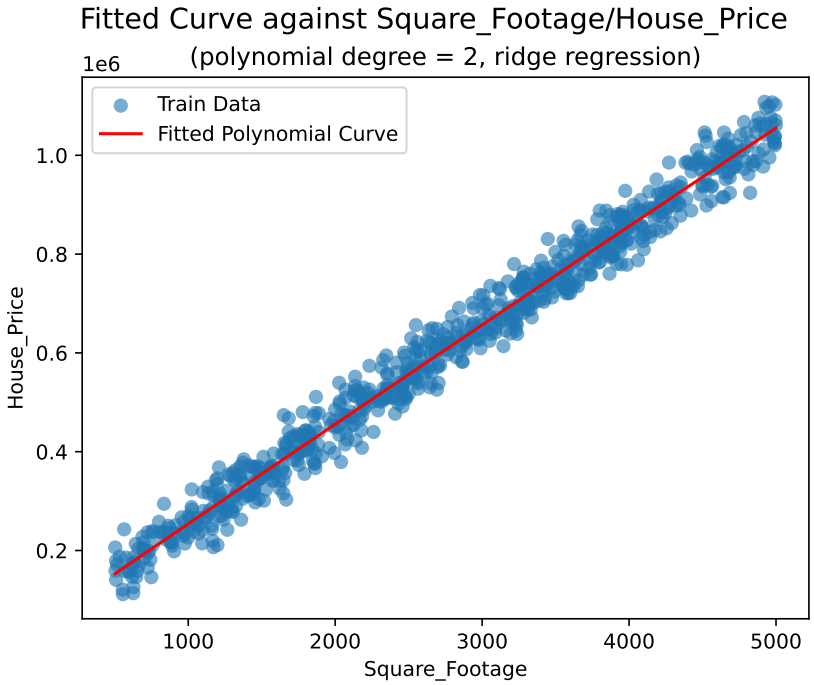


Figure 7: "Square Footage" Degree 2 Polynomial Simple Model



5 Conclusion

Ultimately, the best model was the basic multiple regression linear model with the square footage feature accounting for most of the variation explained by the models created overall. The basic linear model's fit line also corresponded best with the bivariate relationship between the square footage and house price across data objects where the polynomial models suggested inflection points across the range of square footage that is not seen in the data. This shows that in some cases simpler can be better when creating models. While all models were successful, in all categories the, basic linear model served as the best option.